

Notebook: k_means/trial.ipynb

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```
import pandas as pd
```

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```
import numpy as np
```

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```
from sklearn.datasets import load_iris
```

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```
data_array = load_iris()  
data_array
```

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```
data = pd.read_csv('iris.data.csv')
```

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```
def euclidean_distance(x1, x2):  
    """  
    Computes Euclidean distance between two vectors  
    """  
    return np.sqrt(np.sum((x1 - x2) ** 2))
```

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```
x = data.iloc[:150, :-1].values  
x = np.array(x)
```

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```
a, b = x[[1, 2]]
```

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```
a
```

[--- CODE CELL ---]

```
b
```

[--- CODE CELL ---]

```
n, m = data.shape  
# k = n*m  
n, m
```

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```
euclidean_distance(a, b)
```

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```
# random_index = np.random.choice(len(x), n)  
# centroids = x[random_index]  
random_index = np.random.choice(len(x), n)  
cg0, cg1, cg2 = x[random_index], x[random_index], x[random_index]
```

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```

distance0 = []
distance1 = []
distance2 = []

for i in range(n):
    dist0 = euclidean_distance(x[i], cg0[0])
    distance0.append(dist0)

for i in range(n):
    dist = euclidean_distance(x[i], cg1[1])
    distance1.append(dist)

for i in range(n):
    dist0 = euclidean_distance(x[i], cg2[2])
    distance2.append(dist0)
# dist1 = euclidean_distance(x[i], centroids[1])
# dist2 = euclidean_distance(x[i], centroids[2])
# distance1.append(dist1)
# distance1.append(dist2)

```

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```

distance0[0], distance1[0], distance2[0]

```

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```

max(distance0[0], distance1[0], distance2[0])

```

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```

labels = []
for i in range(n):
    labels.append(int((min(distance0[i], distance1[i], distance2[i]))))

```

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```

labels[0]

```

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```

cl0, cl1, cl2 = [], [], []

```

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```

label = np.array(labels)

```

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```

cl0 = x[np.where(label == 0)[0]]
# clusterindex = np.where(label == 0)[0]
# cl0 = clusterindex

```

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```

cl1 = x[np.where(label == 1)[0]]

```

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```

cl2 = x[np.where(label == 2)[0]]

```

[--- CODE CELL ---]

```
cl0
```

```
[ --- CODE CELL --- ]
```

```
cl1
```

```
[ --- CODE CELL --- ]
```

```
cl2
```

```
[ --- CODE CELL --- ]
```

```
cg0 = np.mean(cl0,axis=0)
cg0
```

```
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```

```
cg1 = np.mean(cl0,axis=0)
cg1
```

```
[ --- CODE CELL --- ]
```

```
cg2 = np.mean(cl0,axis=0)
cg2
```

```
[ --- CODE CELL --- ]
```

```
for k in range(10):
    cl0_k, cl1_k, cl2 = []
    print("iteration:",k)
    distance0 = []
    distance1 = []
    distance2 = []
    for i in range(n):
        dist0 = euclidean_distance(x[i], cg0[0])
        distance0.append(dist0)
    for i in range(n):
        dist = euclidean_distance(x[i], cg1[1])
        distance1.append(dist)
    for i in range(n):
        dist0 = euclidean_distance(x[i], cg2[2])
        distance2.append(dist0)
    labels = []
    for i in range(n):
        labels.append(int((min(distance0[i],distance1[i],distance2[i]))))
    label = np.array(labels)
    print(label)
    cl0 = x[np.where(label == 0)[0]]
    cl1 = x[np.where(label == 1)[0]]
    cl2 = x[np.where(label == 2)[0]]
```

```
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```